

Review Analysis System

Team #30


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PROBLEM STATEMENT

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☐ Don't use for recommendations

Our Problem statement is divided into two parts :

1. **Review** - For all the products in the market , there are thousands of reviews available in the market - some positive, some negative and some suggestive. These reviews need to be analysed carefully in order to improve the product and the customer service as well as optimise the profit but doing it manually is not possible with this much amount of data available. Hence, we plan to build a ML model that analyses all the available reviews , suggest changes in the product and also change in prices of the product based on demand and liking,.(the model may also be expanded to getting key insights from an article, essays etc.)
2. **Recommendation** - With so many e- commerce sites coming up, it is important to have a mechanism which suggests products to the users based on their history/cart taking in consideration the type of product that they have bought, range of prices etc. and hence we plan to build an independent recommendation system that solves this problem.

Problem Definition

	Recommendation Part	Review Part
Customer Requirement	B2B: Maximising the sale/revenue for ecommerce players based on user's transactions	B2B: Provide insights to sellers on the products sold based on what their users are saying
Market Survey for available solutions	Amazon's product and Netflix's or Amazon Prime's video recommendation engine	Amazon's review summary (major keywords used wordmap)
Key differentiator & USP	Providing recommendation for products right after a product is added in cart & not checkout	Suggestions for product/service improvement and real time changes in price to sellers based on reviews analyzed in real-time
Business case & barrier to entry	Selling custom trained model to ecommerce players who are in need of improving their sale and or product/service quality based on customer reviews and recommendation model to new e commerce sites	

Technology Landscape Assessment

01	Patents	• Sentiment detection as a ranking signal for reviewable entities Assignee: Google LLC • Nlp-based sentiment analysis Assignee: Fiver LLC
02	Published Literatures	• A Literature Review on Recommender Systems Algorithms, Techniques and Evaluations • A Literature Survey On Recommendation System Based On Sentimental Analysis
03	Open Libraries	• pandas, numpy, matplotlib, seaborn • nltk, textblob, wordcloud, etc
04	Proprietary Libraries	• None

		ROLES	Saurabh	Sanjay	Vedant	Ashwin M	Ashwin S	
Deliverable or Task	Status	Team Members						
RECOMMENDATION PART								
Data Aggregation	100%	R	I	I	A	C		
Data Preprocessing	40%	A	I	I	R	C		
Model Building	0%	R	I	C	S	I		
Interpretation and improvements	0%	D	I	C	R	I		
REVIEW PART								
Data Aggregation	100%	I	S	D	I	R		
Data Preprocessing	40%	I	D	A	I	R		
Model Building	0%	I	R	R	I	I		
Interpretation and improvements	0%	I	D	R	C	I		
D	Driver	Assists those who are responsible for a task.						
R	Responsible	Assigned to complete the task or deliverable.						
A	Accountable	Has final decision-making authority and accountability for completion. Only 1 per task.						
S	Support	Provides support during implementation.						
C	Consulted	An adviser, stakeholder, or subject matter expert who is consulted before a decision or action.						
I	Informed	Must be informed after a decision or action.						

How does the recommendation solution work?

1. Recommendation problem is an unsupervised machine learning problem wherein we have with us the user history i.e. the type of products that the users have bought and analysing that data we recommend similar products to the users.
2. So the code written by us will actually go at the backend of e-commerce website/app , video streaming platforms etc

Flow of the code :

1. We first make an array with each customer's user history.
2. We have product description in the form of words so using a model we convert it to numerical form of vectors
3. We average out all the vectors in a customer's user history and using similarity measures find most similar vectors and thus products

Language /Model/Similarity measure selection

1. Python was used for writing the code.
2. To find similarity between vectors, *cosine similarity* is chosen as it was a good option.
3. To convert 2 layered, *neural network* was thought to be appropriate. But similar task can be performed by *word2vec* model from *gensim* library.

Model in brief :

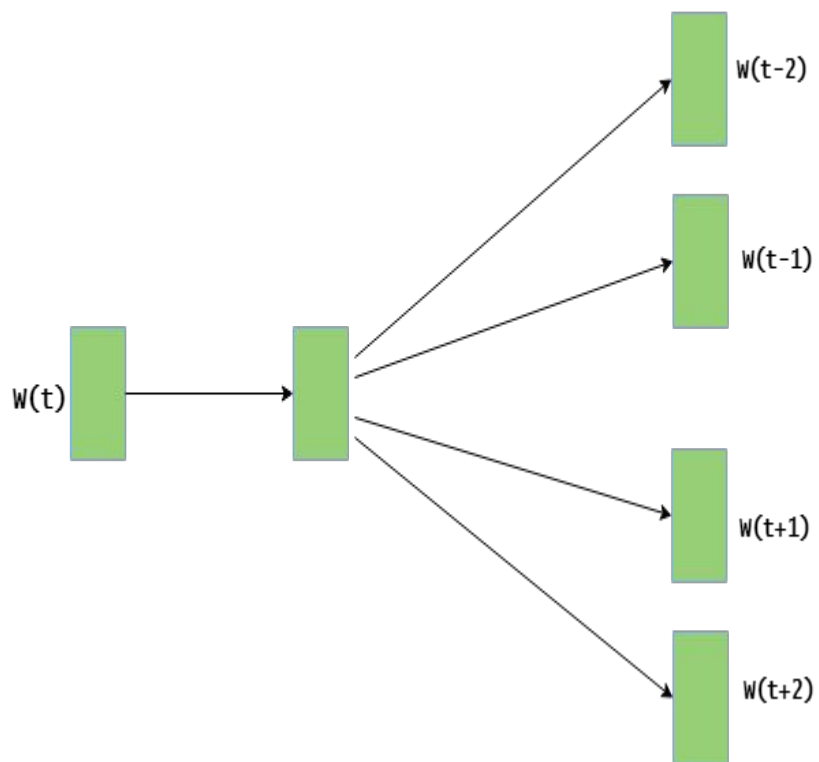
Word2Vec consists of models for generating word embedding. These models are shallow two layer neural networks having one input layer, one hidden layer and one output layer. Word2Vec utilizes two architectures :

1. **CBOW (Continuous Bag of Words)** : CBOW model predicts the current word given context words within specific window. The input layer contains the context words and the output layer contains the current word. The hidden layer contains the number of dimensions in which we want to represent current word present at the output layer.
2. **Skip Gram** : Skip gram predicts the surrounding context words within specific window given current word. The input layer contains the current word and the output layer contains the context words. The hidden layer contains the number of dimensions in which we want to represent current word present at the input layer.

Input

Projection

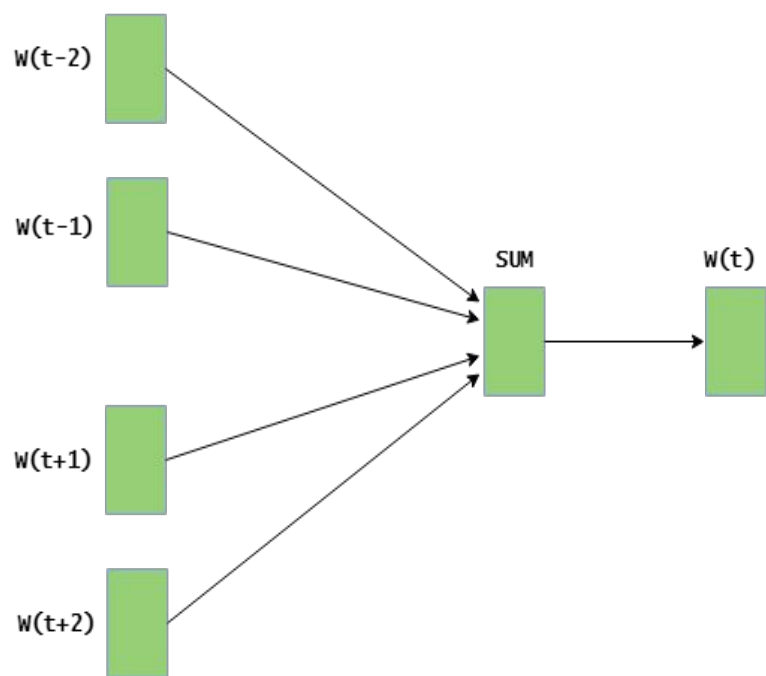
Output



Input

Projection

Output



User interface/ Testing/ Datasets :

- Since it's a B2B product which will be directly merged in the backend of the whole code, having a user interface was of no particular use.
- Since it's an unsupervised learning problem, we cannot test it to find the accuracy. But to demonstrate our code we used an online retail dataset (whose drive link is provided) where in when you run the code you need to enter the customer id and it will return the recommended items.

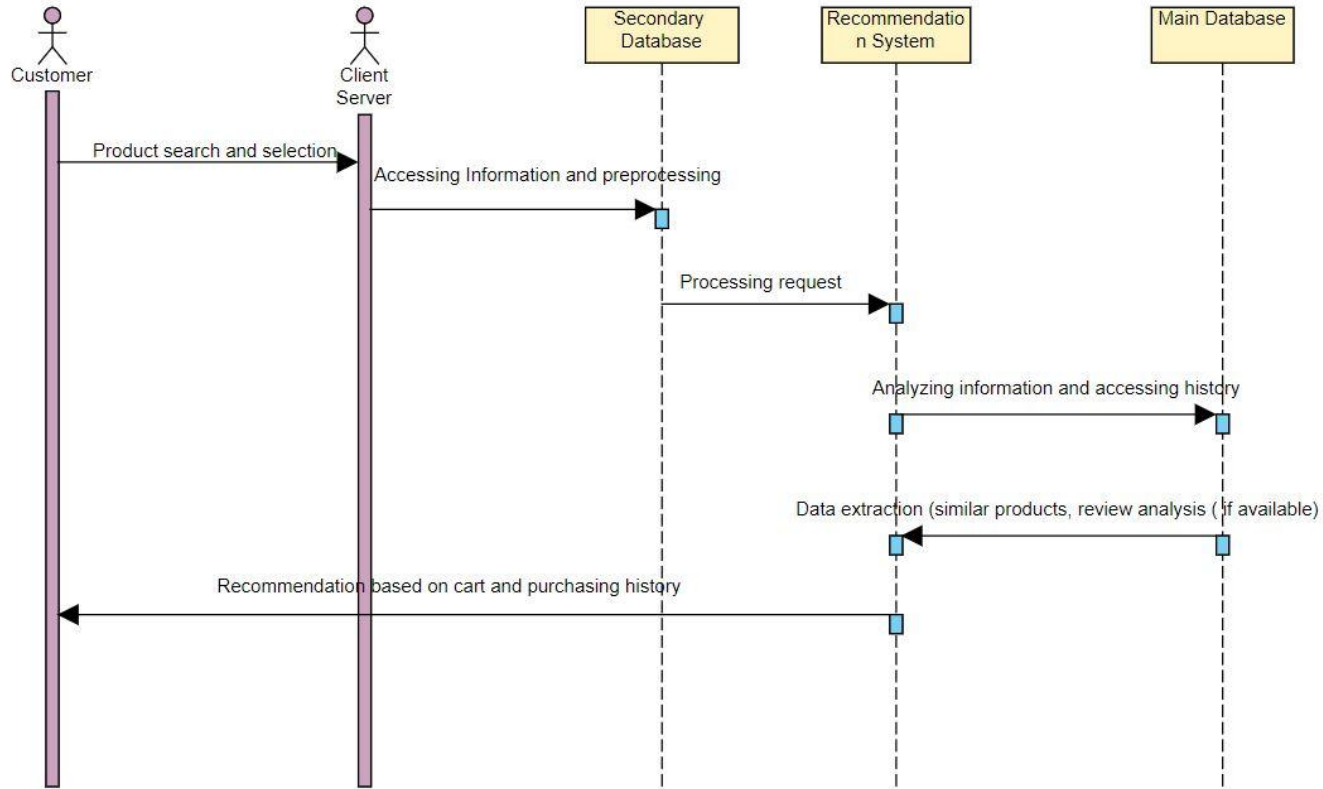
Dataset:

We chose very large electronics item review dataset and its metadata from Amazon to demonstrate the review code.

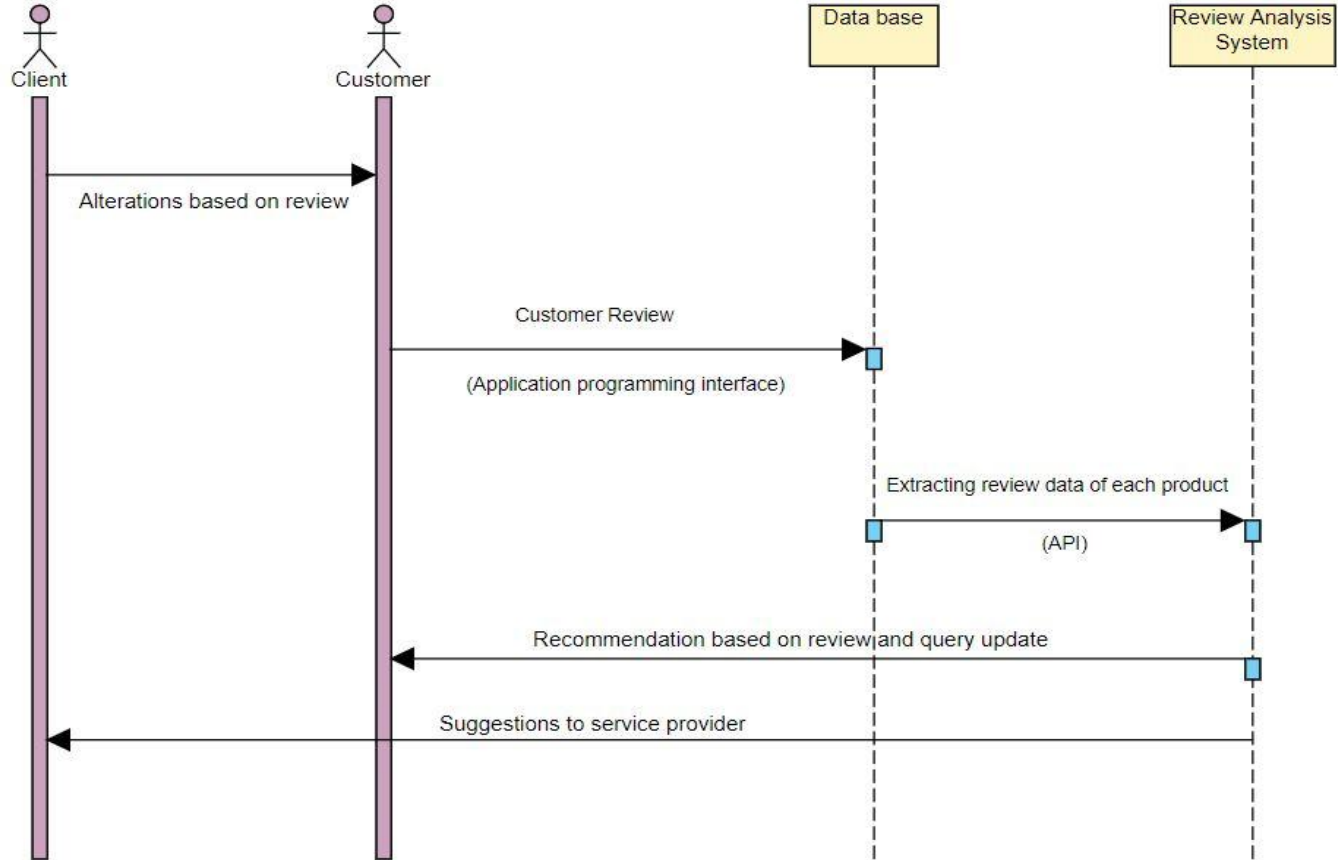
Prospective addition :

An idea of real time management was left as a thought experiment where in each time a new review comes we add it to the database and again run the model and give newer insights of review and recommendation .

Sequence Diagram - Recommendation system



Sequence Diagram - Review system



Brochure

Features

-  Relevant Product Recommendations
-  Review Filtering & Summary
-  Real time price adjustment
-  Maximize Revenue

Best in class in current market



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Review Analysis System

More Data
More Possibilities



Thank You